

Numerical Analysis Exercise 10

Problem 1: Use composite trapezoidal rule

$$\int_a^b f(x)dx = \frac{h}{2} [f(a) + 2 \sum_{k=1}^{n-1} f(x_k) + f(b)] = T(f, h), \text{ where } h = \frac{b-a}{n}.$$

to get the numerical approximation of $\int_0^1 \frac{1}{1+x^2} dx$ and the numerical approximation of π .

Problem 2: Use composite Simpson rule

$$\int_a^b f(x)dx = \frac{h}{3} [f(a) + 2 \sum_{k=1}^{M-1} f(x_{2k}) + 4 \sum_{k=1}^M f(x_{2k-1}) + f(b)] = S(f, h)$$

where $h = \frac{b-a}{2M}$, $k = 0, 1, 2, \dots, 2M$

to get the numerical approximation of $\int_0^1 \frac{1}{1+x^2} dx$ and the numerical approximation of π .

1.

```

1  clc
2  clear all
3  close all
4
5
6  for j = 1:5
7      N = input("请输入切分梯形个数:");
8      formatSpec = "切分梯形个数为:%d";
9      fprintf(formatSpec,N)
10
11     T = CT(0,1,N);
12
13     %%求积分得1/(1+x^2)
14     pi = 4*T;
15     pi = vpa(pi,12)
16 end
17
18 function T = CT(a,b,N)
19     h = (b-a)/N;
20     x = a:h:b;
21     y = 1./(1+x.^2);
22     T = y(1)+y(N);
23     for i = 2:N-1
24         T = T + 2*y(i);

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25 end
26 T = T*(h/2);
27 end
28

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1 切分梯形个数为:1000
2 pi = 3.13959148642
3 切分梯形个数为:10000
4 pi = 3.14139264192
5 切分梯形个数为:100000
6 pi = 3.14157265347
7 切分梯形个数为:1000000
8 pi = 3.14159065359
9 切分梯形个数为:10000000
10 pi = 3.14159245359

```

2.

```

1 clc
2 clear all
3 close all
4
5 for j = 1:5
6 M = input("请输入M:");
7 formatSpec = "切分梯形个数为:%d";
8 fprintf(formatSpec,2*M)
9
10 S = SP(0,1,M);
11 %%对1/(1+x^2)从0到1积分得为4/pi,故pi等于四倍的积分值
12 pi = 4*S;
13 pi = vpa(pi,12)
14 end
15
16 function S = SP(a,b,M)
17 h = (b-a)/(2*M);
18 x = a:h:b;
19 y = 1./(1+x.^2);
20 S = y(1)+y(2*M);
21
22 for i = 1:M-1
23     S = S + 2*y(2*i)+4*y(2*i-1);
24 end
25 S = S + 4*y(2*M-1);
26 S = S*(h/3);
27
28 end
29
30

```

```
1 切分梯形个数为:2000
2  pi = 3.14259240355
3 切分梯形个数为:20000
4  pi = 3.14169265109
5 切分梯形个数为:200000
6  pi = 3.14160265356
7 切分梯形个数为:2000000
8  pi = 3.14159365359
9 切分梯形个数为:20000000
10 pi = 3.14159275359
```